KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 2nd Term Examination, 2015
Department of Computer Science and Engineering
CSE 4211
Machine Learning

TIME: 3 hours FULL MARKS: 210

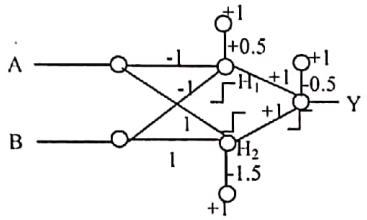
N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

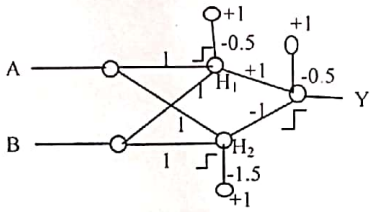
(Answer **ANY THREE** questions from this section in Script A)

1. a) Explain McCulloch-Pitts model of artificial neuron with appropriate sketch. Describe significance of its individual components. (09)
   b) Describe importance of bias weight in artificial neuron. "Only changing the weight value of bias operation of a neuron may change"-justify the statement for appropriate logic gates. (09)
   c) Draw single neuron model for following logics: (08)
   i) $y = x_1' x_2'$, ii) $y = x_1' x_2$, iii) $y = x_1 x_2$, and iv) $y = (x_1 x_2)'$.
   d) Describe shortly the three fundamental and practical issues of learning theory. (09)

2. a) Look the following two neural networks model. (10)



(i)



(ii)

For the model tabulate the outputs of H_1, H_2 and Y for inputs (A and B) values 0 or 1. Are the model performs any logic gate operation? Which ones?
   b) What is the condition to use a function as activation function in ANN? Graphically represents two different forms sigmoidal functions and their derivatives. How derivatives indicate the stability of ANN training? (10)
   c) What is the naming significance of "Back Propagation (BP)" algorithm? Describe basic steps of BP. (07)
   d) What is local gradient in BP training procedure? For an ANN with two hidden layers, write local gradients of each layer. (08)

3. a) What is the effect of learning rate in BP? How momentum parameters speed up training in BP? (05)
   b) What are basic differences between Radial Basic Function (RBF) network and MLP? (05)
   c) Why hybrid training algorithm is required for training RBF networks? Draw sketch diagram of a RBF network and put parameter values to solve XOR problem. (08)
   d) What are the properties and usefulness of the followings: (07)
   i) Local Addressable Memory, ii) Content Addressable Memory, and iii) Associative Memory.
   e) Calculate the weight set (w) of a Hetero- Associative Memory for the following samples: (10)

Pattern	Source, s(p)	Target, t(p)
1	(1 0 0 0)	(1 0)
2	(1 1 0 0)	(1 0)
3	(0 0 0 1)	(0 1)
4	(0 0 1 1)	(0 1)

Finally verify that the system work correctly.

4. a) Differentiate learning and generalization. Which one is more desirable and why? (05)
   b) Why ensemble of classifiers is necessary when a MLP able to capture any arbitrary function? What is the condition for better ensemble construction? (05)
   c) Among the studied ensemble methods, whose are able to take facility of multi-processing system? How? (07)

d) Describe an ensemble method which promotes diversity through artificial patterns. (08)
e) What is support vectors in support vector machine (SVM)? How SVM may classify non linearly separable data? Explain with appropriate examples. (10)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Why we prefer nature-inspired heuristic methods although they might not give exact solution for a particular optimization task? (05)
b) Why selection is necessary in Genetic Algorithm (GA)? Write down the differences between Rank based selection and Tournament selection. (12)
c) Write down the importance of using mutation in GA. Also write down the effects of mutation rate in GA. (07)
d) Is it possible to find out the population size from the Roulette wheel given below? Explain. (06)



Fig. 5(d)

e) What is Elitism? Why is it important in GA? (05)

6. a) "In Particle Swarm Optimization (PSO) every particle updates position in each and every iteration even it is the global best"-justify the statement from the velocity and position update equations. (08)
b) Write down the working procedure of Attractive Repulsive PSO and Multiswarm PSO. (08)
c) What is the effect of Lifespan controller in ALC-PSO? How Lifespan of a leader is adaptively tuned? Explain. (12)
d) Write down the similarities and dissimilarities between GA and PSO. (07)

7. a) Describe the importance of transition probability in selecting the next city in TSP using Ant Colony Optimization (ACO). (10)
b) "Pheromone evaporation pushes ACO convergence"-is it true? Justify your opinion. (08)
c) Explain the Employed bee phase, onlooker bee phase and scout bee phase of Artificial Bee Colony Optimization (ABCO) algorithm. (12)
d) Explain differences of Biogeography-Based Optimization (BBO) from ACO and PSO. (05)

8. a) How does Cuckoo Search (CS) algorithm provide exploration and exploitation? Explain. (08)
b) How does Bacterial Foraging Optimization Algorithm (BFOA) find the optimum solution? (05)
c) What is the benefit of using OPTICS clustering algorithm? How this algorithm works? Explain. (08)
d) Define the core point and border point. Consider the data points in the following figure Fig. 8(d) with radius where Minpts = 6. Find out the core points and border points. (06)

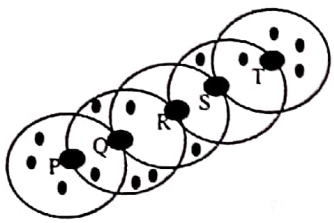


Fig. 8(d)

Is P and T density reachable from each other?

e) Consider a set of records that have three attributes namely salary, age and vacation. How CLIQUE algorithm clusters data objects? Explain. (08)